Question:
An achromatic prism combination is to be constructed of crown- and flint-glass components For light of wavelengths 6563 \AA, 5893 \AA, and 4861 \AA, the crown glass has relative indices of refraction of 1.5146, 1.5171, and 1.5233, respectively and the relative indices of refraction of the flint glass are 1.6224, 1.6272, 1.6385, respectively. The wavelengths of 6563 \AA, 5893 \AA, and 4861 \AA, correspond to red, yellow, and blue light, respectively. The crown-glass prism to be used has an angle of 10°. What must be the angle of the flint-glass prism?
Answer Choices:
A. 10 degrees
B. 7.20 degrees
C. 4.45 degrees
D. 3.75 degrees
E. 2.50 degrees
F. 5.40 degrees
G. 9.00 degrees
H. 8.15 degrees
I. 6.30 degrees
J. 11.00 degrees

Reasoning + Answer:

</tool_call> reasoning process here
For an achromatic prism combination, the condition for dispersion cancellation (i.e., the deviation of red and violet light must be equal and opposite) is used. The net angular dispersion must be zero for the combination of crown and flint glass prisms.

The angular deviation of light through a prism is approximately given by:

Î´ = (n - 1) A

where:
- Î´ is the angular deviation,
- n is the refractive index of the glass,
- A is the prism angle.

For achromatic combination, the angular dispersion (which is proportional to the derivative of deviation with respect to wavelength) must cancel out. The condition is:

Î´_red + Î´_blue = 0

This leads to the condition:

(n_crown - 1) A_crown + (n_flint - 1) A_flint = 0

But more precisely, for achromatic combination, the dispersion condition is:

(ÎĶn_crown) ĂĹ A_crown = (ÎĶn_flint) ĂĹ A_flint

where ÎĶn is the variation of refractive index with wavelength (i.e., the change in n over the wavelengths).

We are given:

For crown glass:
- Î» = 6563 Ãħ (red): n = 1.5146
- Î» = 5893 Ãħ (yellow): n = 1.5171
- Î» = 4861 Ãħ (blue): n = 1.5233

So, the change in refractive index for crown glass:
- ÎĶn_crown = n_blue - n_red = 1.5233 - 1.5146 = 0.0087

For flint glass:
- Î» = 6563 Ãħ: n = 1.6224
- Î» = 5893 Ãħ: n = 1.6272
- Î» = 4861 Ãħ: n = 1.6385

So, ÎĶn_flint = n_blue - n_red = 1.6385 - 1.6224 = 0.0161

The condition for achromatic combination is:

(ÎĶn_crown) ĂĹ A_crown = (ÎĶn_flint) ĂĹ A_flint

So:

0.0087 ĂĹ 10Â° = 0.0161 ĂĹ A_flint

Solve for A_flint:

A_flint = (0.0087 ĂĹ 10) / 0.0161 = 0.087 / 0.0161 âīĪ 5.404

So, A_flint âīĪ 5.40 degrees

Therefore, the angle of the flint-glass prism must be approximately 5.40 degrees.

<answer> F </answer>